

upper-left box in figure 7.1) positing senses for unknown words, hypothesizing multiword expression analyses, using selectional constraints to infer the meaning of unknown words, using learning-oriented preferences to disambiguate elements of input, and resolving incongruities in meaning representations by creating new senses of known words.

- **Text Meaning Representation.** The output of Natural Language Understanding is a Text Meaning Representation (TMR) that reflects the results of learning-oriented reasoning. The TMR includes (a) the most specific semantic analysis of newly learned words and phrases that the agent is able to formulate using the methods available so far, and (b) semantic analyses of whatever new world knowledge might have been presented, which will be incorporated into the ontology during ontology learning. For example, if a language input is *Endoscopy is a very low-risk procedure*, the TMR will represent that meaning as “ENDOSCOPY (RISK .1).” During ontology learning, the agent will use this meaning representation to add the property value “RISK .1” to the concept ENDOSCOPY.
- **Reason about Learning.** Most scripts in this module involve reasoning in service of learning ontological concepts and their properties. However, certain lexicon learning procedures are carried out as well, such as remapping tentative new word senses to newly learned ontological concepts.
- **Select Learning Move.** Finally, the agent must decide what to do with the candidate knowledge. There are five options. The first one, record the knowledge, is selected in optimal cases, when the candidate knowledge is of high quality and high confidence. In suboptimal cases—when something is underspecified or missing—the agent chooses among the other four options: wait and see if the needed information is forthcoming; ask a human a question about it; launch data-driven learning to try to find the needed information; or cancel the learning of the entity, deeming it unnecessary within the given application.

Below is an example of an input that the LEIA can process well enough to directly record the candidate knowledge in its lexicon and ontology. In this example, the LEIA is being trained in the medical domain by a human who explains:

Systemic sclerosis is a multisystemic autoimmune disease of unknown origin that affects connective tissue. It often affects the esophagus and can cause heartburn, regurgitation, dysphagia, and chest pain.

To understand how the agent processes this input, it is important to understand the initial state of its lexicon. It knows all of the words except for *systemic* and *sclerosis*; it knows the multiword expressions *autoimmune disease*, *of unknown origin*, and *connective tissue*; and it has a construction that anticipates nominal compounds of the form “Noun_{BODY-PART} *pain*,” which enables the precise analysis of *chest pain* as “PAIN (LOCATION CHEST).”

Of the ten learning functions shown in figure 7.1—five attached to Natural Language Understanding and five attached to Reason about Learning—six are leveraged in processing this input:

- Posit senses for unknown word: the agent posits senses for the adjective *systemic* and the noun *sclerosis*.
- Hypothesize a multiword analysis: the agent reasons that these words likely form the multiword expression *systemic sclerosis*.
- Disambiguate using learning preferences: the agent disambiguates *is* using a definitional sense that maps to the ontological relation IS-A, and it disambiguates *the* in *the esophagus* by preferring a generic interpretation due to the lack of a coreferential antecedent and the fact that ontology learning is underway.
- Create and anchor new concept: the agent generates the concept SYSTEMIC-SCLEROSIS from the newly learned lexical sense *systemic sclerosis*, and it anchors the new concept in the ontology as a child of AUTOIMMUNE-DISEASE based on the previously established IS-A relation.
- Finalize new lexical sense: The agent remaps the lexical sense *systemic sclerosis* from EVENT, which was the initial mapping during TMR creation, to the new concept SYSTEMIC-SCLEROSIS.
- Learn property values: The agent learns all of the property values presented in the input, as shown in the final concept frame it learns.

SYSTEMIC-SCLEROSIS		; systemic sclerosis
IS-A	AUTOIMMUNE-DISEASE	; is an autoimmune disease
THEME	SET-1	; that affects multiple systems
CAUSED-BY	IDIOPATHIC-EVENT	; it is of unknown origin
EFFECT	CHANGE-EVENT-1	; it affects connective tissue
EFFECT	CHANGE-EVENT-2	; it often affects the esophagus
HAS-TYPICAL-SYMPTOM	SET-2	; it has a set of typical symptoms
SET-1		
MEMBER-TYPE	ANATOMICAL-STRUCTURE	
CARDINALITY	>1	

CHANGE-EVENT-1	
THEME	CONNECTIVE-TISSUE
CHANGE-EVENT-2	
THEME	ESOPHAGUS
FREQ-ACROSS-INSTANCES	.7 ; “often”
SET-2	
ELEMENTS	DYSPHAGIA, PAIN-1, REGURGITATE, HEARTBURN
PAIN-1	
LOCATION	CHEST

This frame looks very similar to the TMR that was produced as a result of language understanding. The key differences are: numerical indices are used differently in TMRs and ontological frames; and whereas the TMR mapped *systemic sclerosis* to the generic EVENT since the agent had not yet learned the concept SYSTEMIC-SCLEROSIS, the ontological frame uses the newly learned concept.

The final stage of learning is Select Learning Move, where the agent evaluates whether its candidate additions to the lexicon and ontology are fit to be recorded to memory. In this example, all of them are.

To reiterate the point of this example: in some cases, language inputs are sufficient to seed learning without the agent needing to ask a human for clarification or consult a text corpus for additional information. It behooves teachers of LEIAs to present information in a clear and organized way, just as teachers strive to do when interacting with human students.

Of course, it will happen that the agent’s analysis of an input is not sufficient to support confident learning: there might be residual ambiguities in the language input, a new term might not have been defined, and so on. In dialog applications, the agent can then ask the human for help. This is called mixed-initiative learning, to which we now turn.

7.1.2 Mixed-Initiative Learning

Mixed-initiative learning is dialog-based learning in which both the teacher and the trainee can initiate moves. In mixed-initiative learning, the big questions are *when*, *why*, and *how* an agent takes the initiative to ask questions. This decision-making is informed by two kinds of parameters: those involving learning per se and those relevant to specific applications.

- **Parameters involving learning per se.** The agent knows basic principles about the minimal requirements for learning lexical and ontological knowledge and, therefore, can detect when candidate knowledge is not